$\therefore N(T) = \{\alpha(-1,1,0)\}$

$T(1,0,0) = (1,1,1)$

$T(0,1,0) = (1,1,1)$

$T(0,0,1) = (1,0,0)$

$R(T) = [\{(1,1,1),(1,0,0)\}]$

Q:-(1) $T(1,0,0) = (0,0,0)$

$T(0,1,0) = (0,1,0)$

$T(0,0,1) = (0,0,1)$

$R(T) = [\{(0,1,0),(0,0,1)\}]$

$= yz$-plane.

* Let $T: U \to V$ be a linear map, then

(i) $R(T)$ is a subspace of V

(ii) $N(T)$ is a subspace of U.

(iii) T is one-one iff $N(T)$ is a zero subspace of U.

(iv) If $[u_1, u_2, \ldots, u_n] = U$, then

$R(T) = [T(u_1), T(u_2), \ldots, T(u_n)]$

(v) If U is finite dimensional, then $\dim R(T) \leq \dim U$.

(iii) Let T is one-one.

i.e. $T(u_1) = T(u_2)$

$\Rightarrow u_1 = u_2$

$\Rightarrow u_1 - u_2 = 0$

$\Rightarrow T(u_1 - u_2) = 0$ } X

Pkalika Maths